

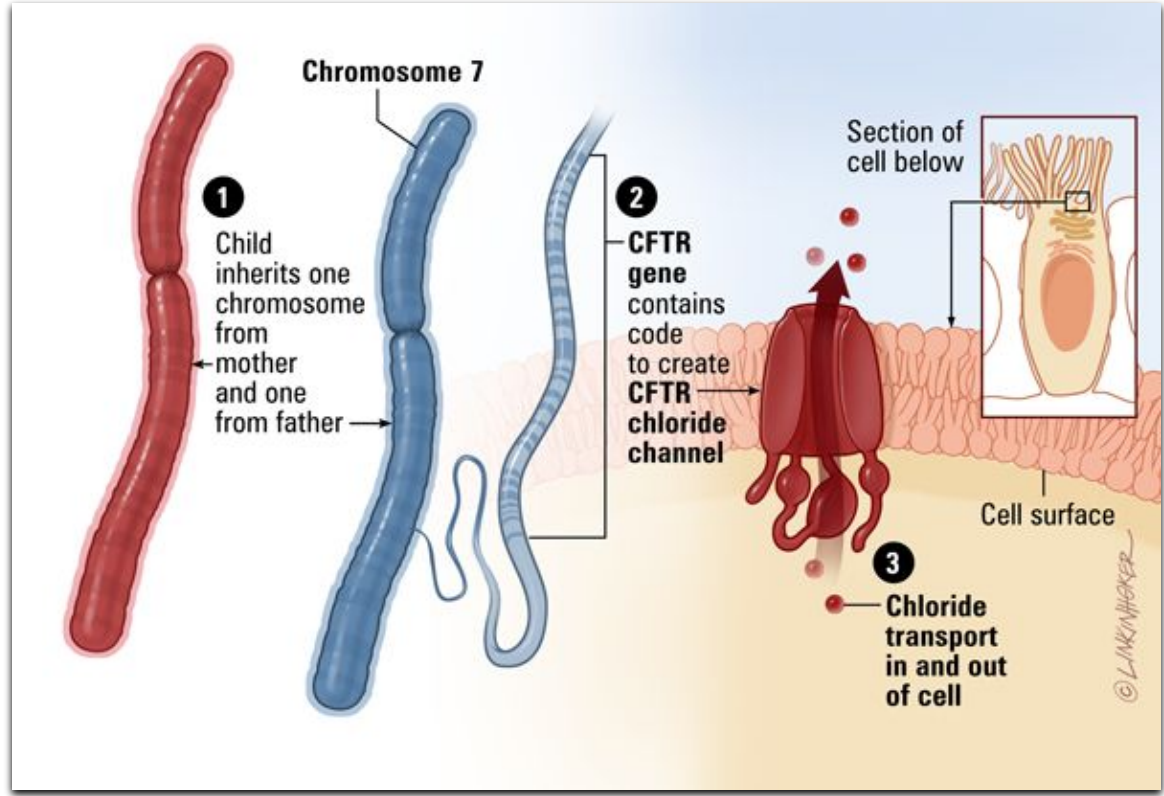
Characterization of a Rare Missense Mutation in the Cystic Fibrosis Transmembrane Conductance Regulator Gene

Diego Aguirre, Hajar Al-Tal, Victoria Arinella, and Marie Koumba

Abstract: Cystic fibrosis (CF) is a genetic disorder caused by mutations in the CFTR gene, which can disrupt chloride transport and lead to thick mucus accumulation in organs such as the lungs and pancreas. In this mock patient treatment case, a 16-year-old male with a preexisting diagnosis of CF presented to the emergency room with difficulty breathing and elevated blood glucose levels, leading to a diagnosis consistent with cystic fibrosis-related diabetes. Since the patient’s primary cells were not directly available, HEK293T adherent epithelial cells were used as a model system to study the patient’s CFTR mutation and evaluate potential modulator treatments. Sequence analysis identified the CFTR variant as ValΔ520Phe. HPLC was then used to compare Drug A/Alyftrek, Drug B/Ivacaftor, and Drug C/Lumacaftor by measuring caffeine remaining in the media as an indicator of drug uptake. Alyftrek showed the strongest result, with no caffeine detected, while Drugs B and C showed little to no improvement compared with the controls. Overall, these findings suggest that Alyftrek may be the most effective modulator tested for this patient’s rare CFTR mutation and support the importance of personalized treatment approaches for cystic fibrosis.

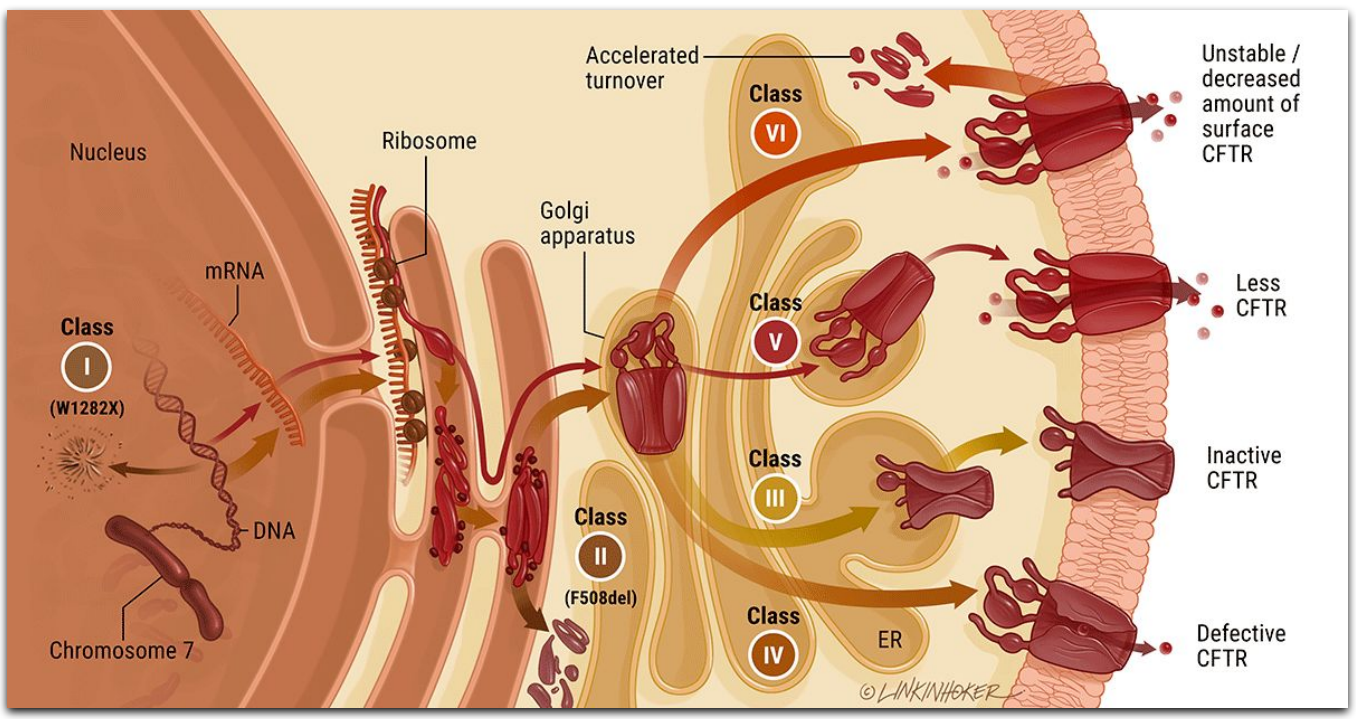
Background:

Cystic Fibrosis (CF) is a rare autosomal genetic disorder characterized by an accumulation of thick mucus in organs including the lungs, pancreas, kidneys, and intestines. CF affects about 1 in 3,000 newborns. Common symptoms include difficulty breathing, frequent lung infections, sinus infections, poor growth, fatty stool, and infertility in most males.



<https://hopkinscf.org/>

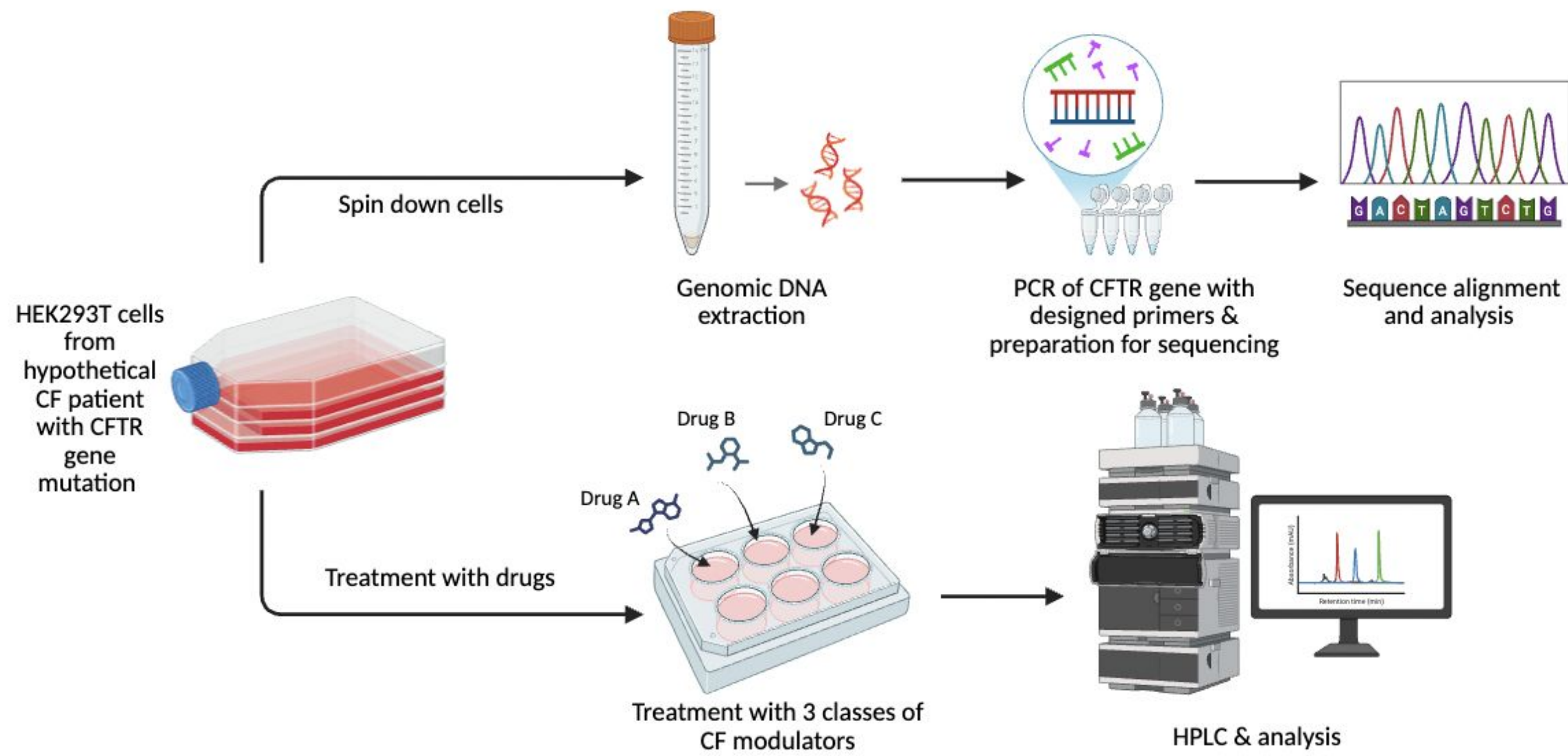
The Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) gene is located on the long arm of chromosome 7 and encodes the CFTR protein. In the lung, the CFTR proteins acts as a chloride channel and is found in the apical membrane of cells lining the airway.



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Mutations in the CFTR gene can affect the CFTR’s synthesis, function, or stability. The six major classes of CFTR mutations are **I**: protein synthesis defect; **II**: misprocessing, misfolding, defective, or trafficking defect; **III**: regulation or gating defect; **IV**: reduced channel conductance; **V**: reduced synthesis/quantity; and **VI**: reduced stability.

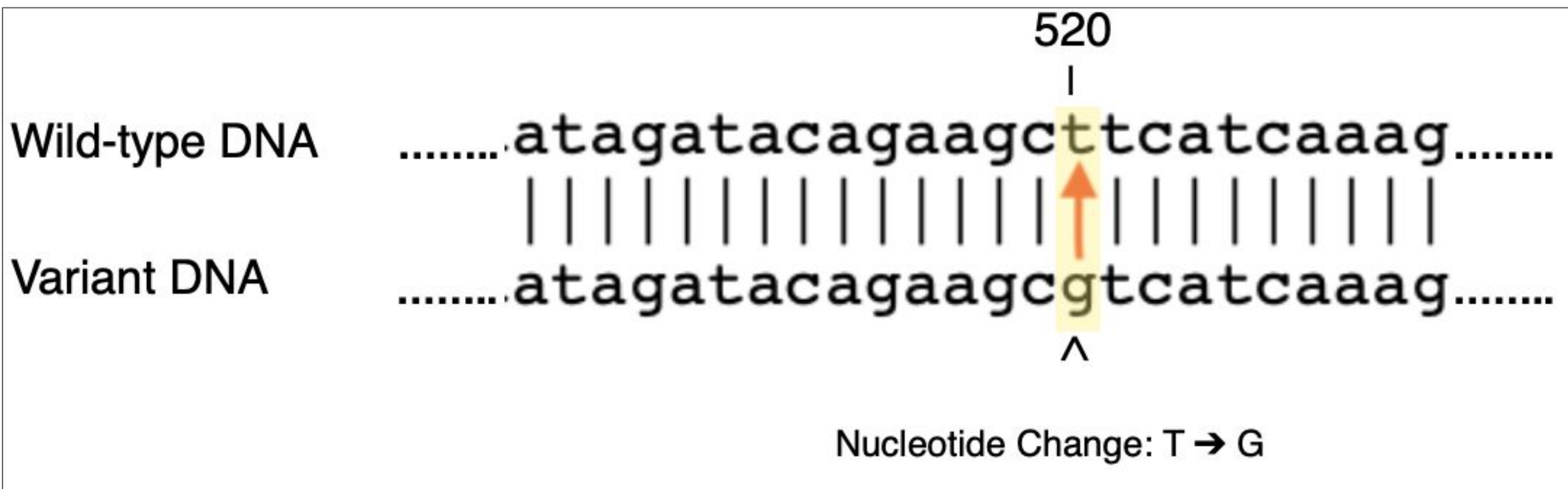
Materials and Methods:



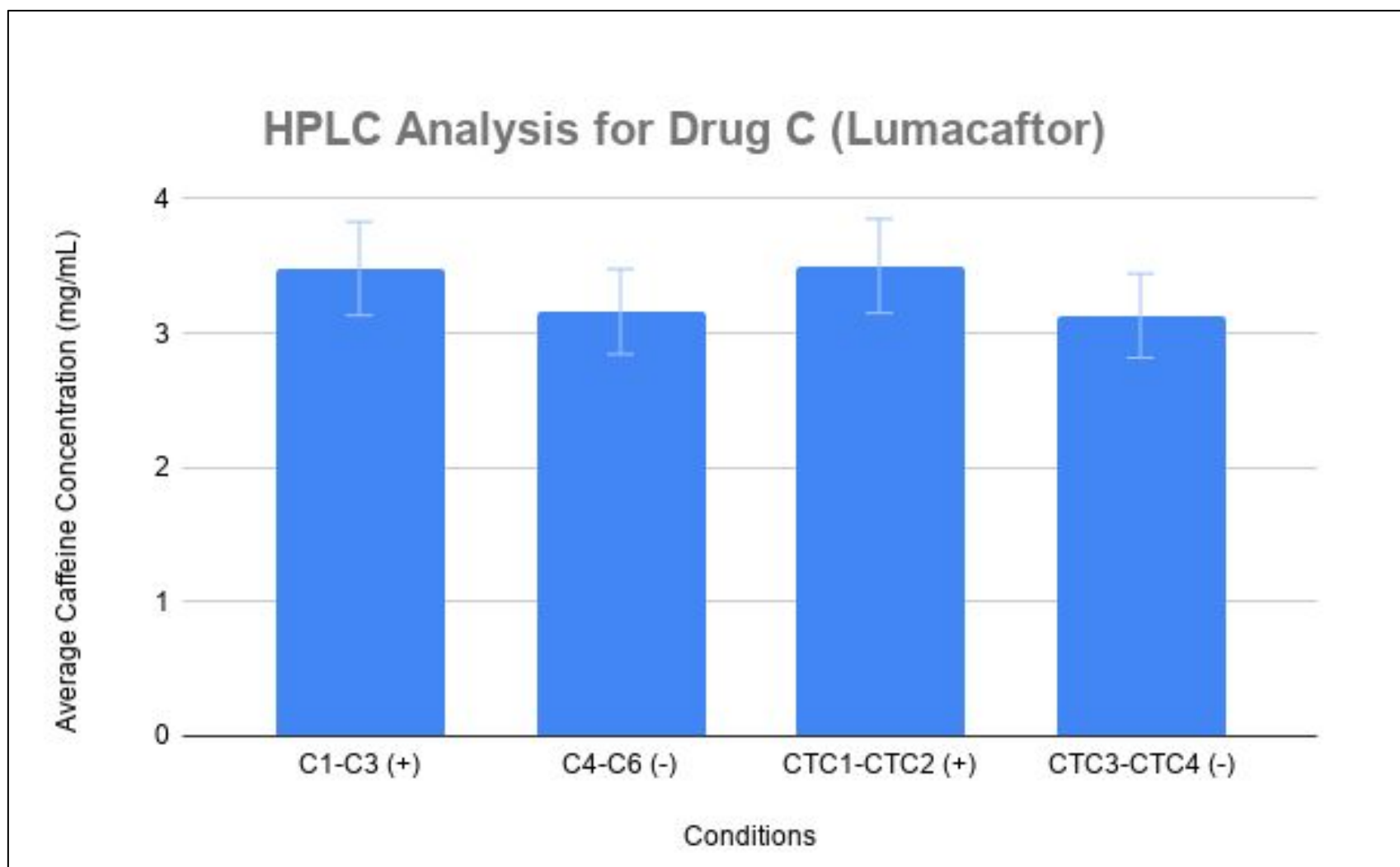
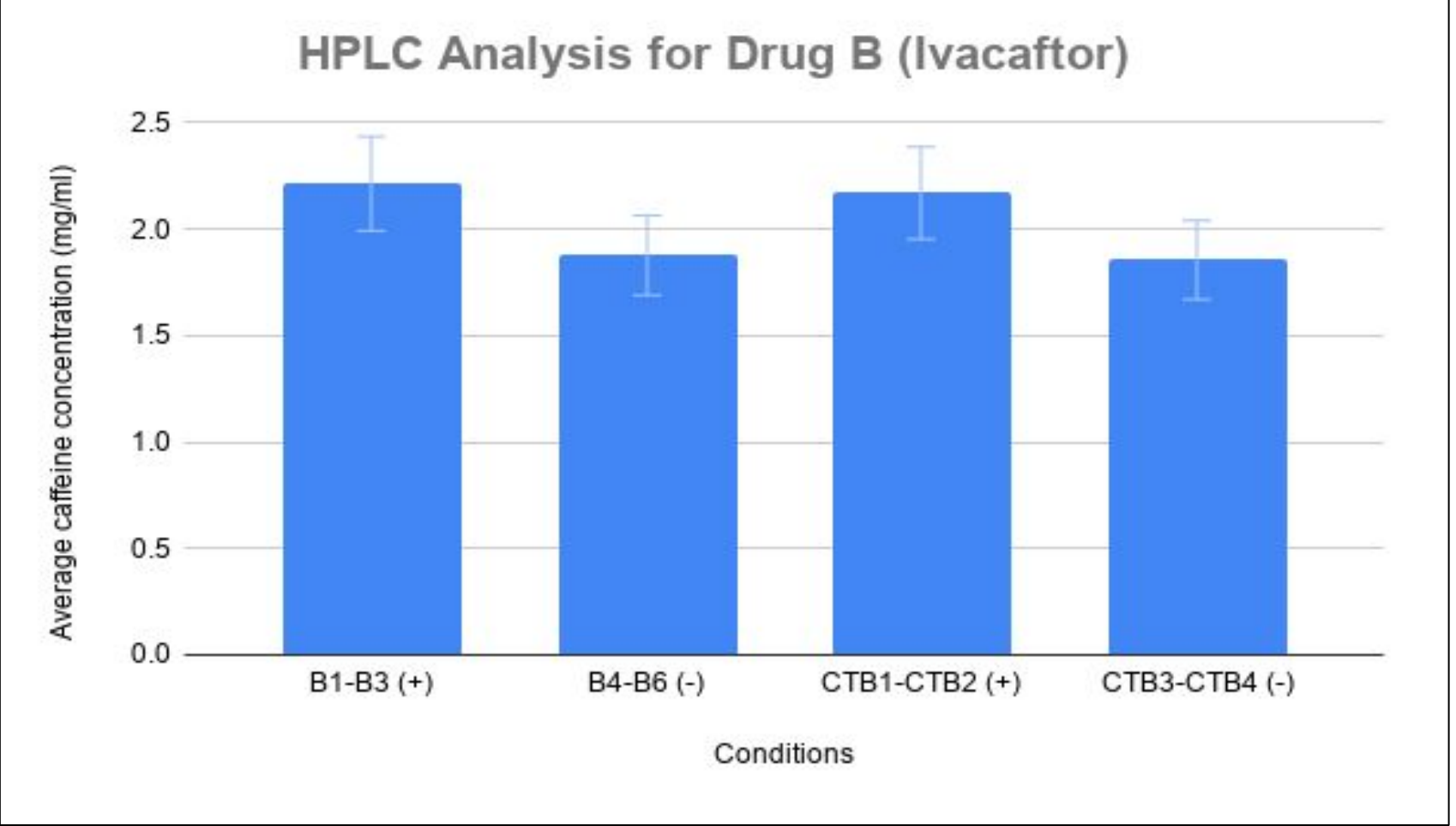
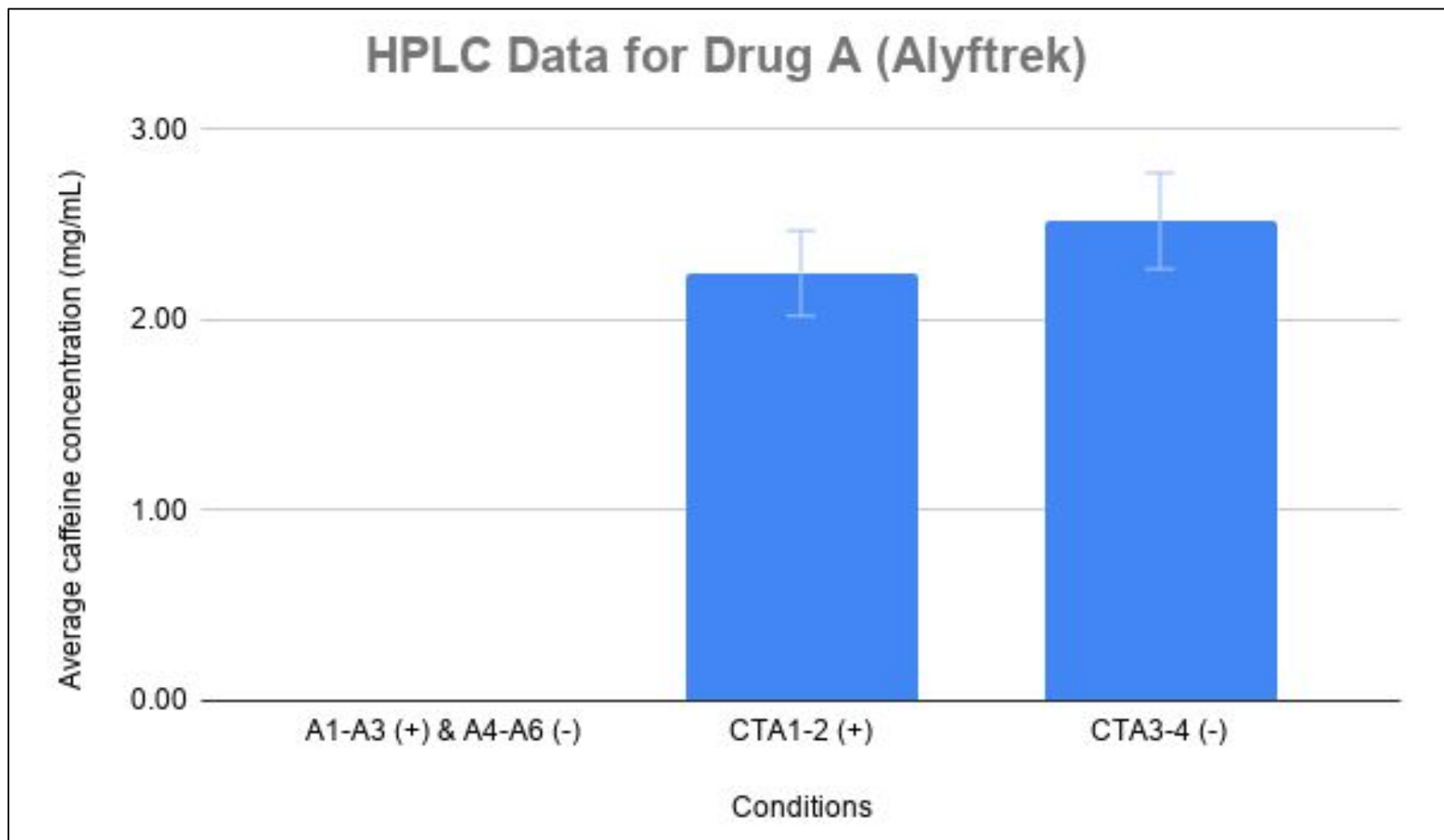
Data and results:

HEK293T cells were maintained in culture and passaged at a 1:2 split ratio after reaching 70–80% confluency. Genomic DNA of sufficient quality was extracted for downstream analysis. Endpoint PCR using primers designed for the CFTR mutation region produced clear bands on agarose gel electrophoresis, indicating successful amplification of the target locus. The amplified product was sequenced, and sequence analysis identified the CFTR variant as ValΔ520Phe. HPLC was then used to compare Drug A (Alyftrek), Drug B (Ivacaftor), and Drug C (Lumacaftor) by measuring caffeine remaining in the media. Among the drugs tested, Drug A showed the strongest result.

Wild-Type and Variant CFTR DNA Sequence Alignment

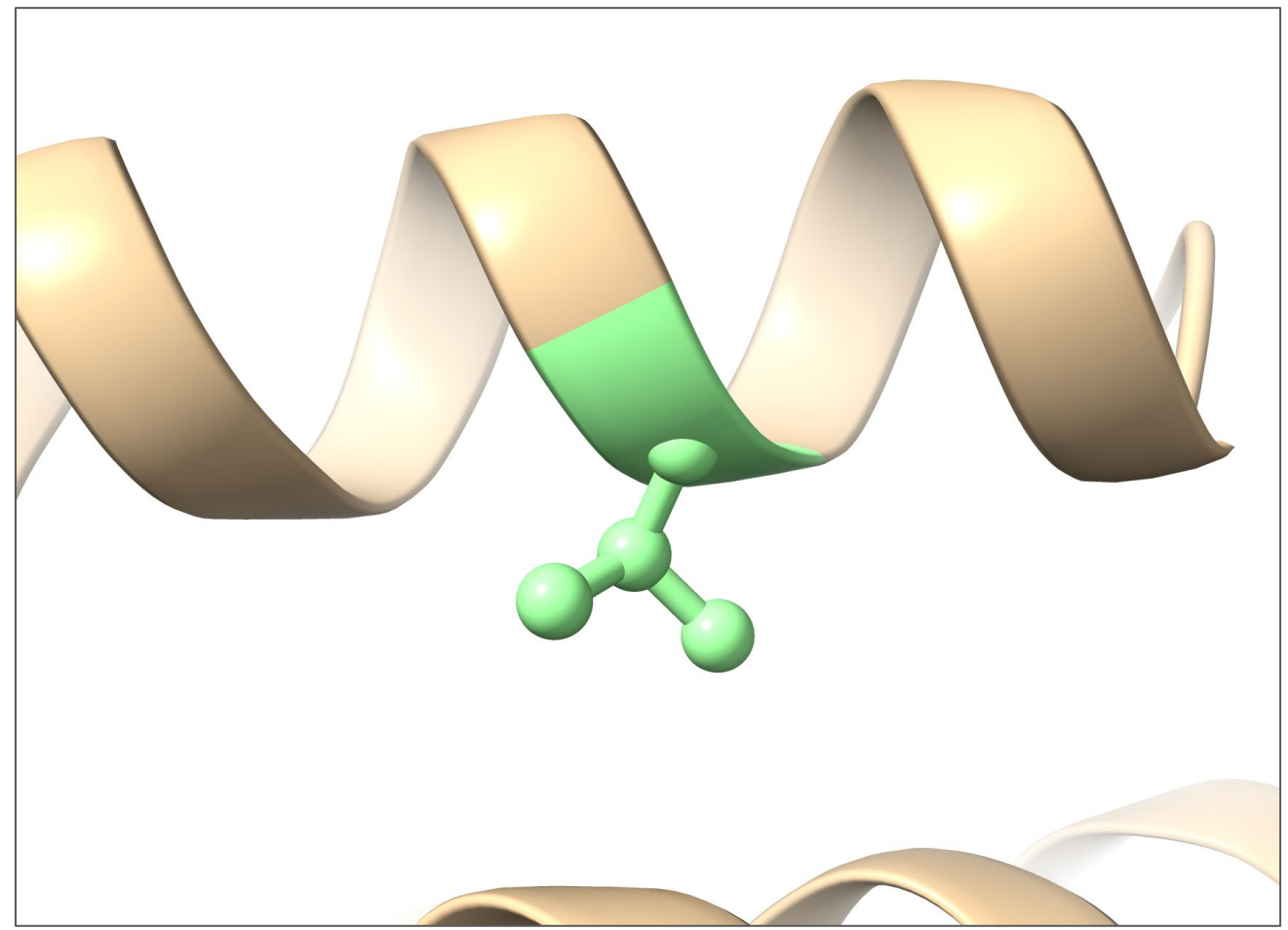


HPLC Analysis of CFTR Modulator Uptake



Discussion:

Sequence analysis identified an uncommon ValΔ520Phe missense mutation in the CFTR gene, located in the NBD1 domain, a region important for ATP binding and regulation of CFTR channel activity. Since this amino acid change may disrupt normal protein function, it likely contributes to the patient’s cystic fibrosis symptoms. Drug screening results showed that Alyftrek had the greatest cellular uptake, indicating it was the most effective modulator tested. In contrast, Ivacaftor and Lumacaftor showed no meaningful uptake. Overall, these findings support Alyftrek as the strongest treatment for a more personalized CF treatment approach. Future studies, such as a Forskolin-Induced Swelling (FIS) assay, this can provide insight into the patient’s CFTR function with Alyftrek.



Modelling done using ChimeraX. This highlights the shift from the smaller, hydrophobic Valine (left/green), to the bigger aromatic and hydrophobic Phenylalanine (right/red) in the NBD1 domain.

Conclusions:

Our results suggest that the patient’s rare ValΔ520Phe CFTR mutation is most responsive to Alyftrek, while Ivacaftor and Lumacaftor showed little to no benefit. Overall, this study highlights the importance of characterizing rare CFTR mutations to support more personalized and effective treatment strategies for cystic fibrosis.

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